

Interface Capability Brief

FHIR Patient REST Demo · Stakeholder overview · 2026-08-05

1. Executive summary

Demo PilotFish accepting HL7 FHIR R4 Patient create/read over HTTP REST with synchronous responses.

2. Who this is for

- Sources: FHIR HTTP clients (Web UI, curl, LAN devices) - Destinations: Sync HTTP response; SQL FhirResources; file mirror under output/fhir-store - Demo vs production: Demo only

- Sources: FHIR HTTP clients (Web UI, curl, LAN devices)
- Destinations: Sync HTTP response; SQL FhirResources; file mirror under output/fhir-store
- Demo vs production: Demo only

3. What the interface can do

Capabilities below are drawn from the working design and, when present, the published CapabilityStatement — not from aspirational roadmap items.

- Transport: RESTfulWebServiceListener — /eip/rest/fhir/{Resource}/{id}]
- Format: application/fhir+json
- Interactions: POST Patient, GET Patient/{id}
- Identity: FHIR id; MRN via identifier.value (demo expects MRN-*)
- Samples: samples/*.json

Integration highlights

Stage	Module	Notes
Listener	com.pilotfish.eip.modules.http.rest.RESTfulWebServiceListener	SERVICE_NAME=fhir, Synchronous=true
Extract / validate	Regex + Attribute Population	Heuristic PASS/FAIL for create
Router	XPathRoutingModule OGNL on method/resource/validation	Create / Read / 400 / 405
Persist	FileWriteProcessor + DatabaseSqlProcessor MERGE	After validation on create
Read	DatabaseSqlProcessor SELECT + XSLT	Empty → OperationOutcome
Status / headers	HttpStatusCodeProcessor, AddHttpHeadersProcessor	fhir+json, Location
Reply	com.pilotfish.eip.modules.internal.SynchronousResponseTransport	Sync body

4. Honest boundaries (not claimed yet)

- Add Bundle transaction / search in a later phase
- Basic auth on REST listener for demos?

5. What you can verify in a walkthrough

Use these as checklist items when reviewing the demo with business stakeholders. They describe outcomes, not module names.

- Exercise route “1 - FHIR Patient REST API” from the Web UI or documented smoke path
- Confirm outputs land in the documented folders / queues and appear in the Web UI status views

6. How it works (plain language)

7. Validation

- Checked: resourceType Patient, id, MRN-ish identifier value, family name - Not checked: full FHIR profiles, terminology, OAuth/SMART - Failure blocks create with HTTP 400 OperationOutcome

6. State & idempotency

- MERGE upsert on create (same id updates RawFhir) - Respond after persist - Dedup: unique (ResourceType, ResourceId)

8. Dual-write / side effects

- Order: file write → SQL MERGE → sync response body restored from attribute - Accepted demo risk: file/SQL dual-write without outbox

3. Inbound contract

- Transport: RESTfulWebServiceListener — /eip/rest/fhir/{Resource}/{id} - Format: application/fhir+json - Interactions: POST Patient, GET Patient/{id} - Identity: FHIR id; MRN via identifier.value (demo expects MRN-) - Samples: samples/.json

Routes in this interface

Each route is a configured PilotFish flow. Technical wiring appears in the Route Diagrams PDF; here is the stakeholder reading.

1 - FHIR Patient REST API: runs steps such as FHIR R4 REST, Save Raw Body, Extract Logical Id, Extract MRN.

7. Security & trust posture

- OAuth2 / OIDC Bearer tokens (Keycloak in this demo) protect write and export operations
- Inbound clinical/FHIR payloads can be validated before persistence
- Credentials and passwords in this package are demo-only — not production secrets
- Risk noted in design: High — REST listener quirks on 23R1 — Smoke-tested with curl
- Risk noted in design: Med — Heuristic validation — Labeled demo-only
- Risk noted in design: Med — Dual-write file+SQL — Accepted demo risk
- Risk noted in design: Low — No search/transaction yet — Scoped in README

8. Seeing it work

- - Ports: SQL 14337, EIP 8102, Web UI 8103 - LAN: LAN_HINT=http://192.168.68.52:8103/; FHIR public base on :8102 - Cold start ~60–90s
- - Local: http://localhost:8103/documents/fhir-rest-research.pdf
- docker compose up -d --build
- - Web UI local: http://localhost:8103/

9. Related technical documents

- DESIGN.md — working engineering specification for this interface

-
- README.md — how to run, ports, and smoke commands
 - FHIR_V2_Route_Diagrams.pdf — detailed PilotFish route wiring diagrams
 - This Capability Brief — shareable stakeholder summary (regenerate after design changes)

This document is generated from the interface sources listed above. If design and runtime diverge, believe the smoke-tested runtime and update DESIGN.md.